ANSWERS DAY 2

SPOT 1

1. 1ST degree heart block. 2. 75 beats/min approx. 3. Tall tented T waves

SPOT 2

1. 1 – 2 → Isovometric contraction; 2 – 3 → Phase of Ejection; 3 – 4 → Isovometric relaxation; 4 – 1 → Phase of Filling
2. Cardiac Output: The volume of blood pumped by each ventricle into their respective artery per minute
   Stroke Volume: The volume of blood pumped by each ventricle into their respective artery per each contraction (per beat)
3. Cardiac index (CI) is a hemodynamic parameter that relates the cardiac output from left ventricle in one minute to body surface area. Its normal range is between 2.5 and 4.2 L/min/m^2. (3.2 is average)

SPOT 3

1. Cardiac conducting cell
2. Phase 0 – Depolarization; Phase 3 – Repolarization; Phase 4 – Prepotential or pacemaker potential
3. Phase 0 – Na+ enters via V_{Na+} channels; Phase 3 – K+ exits via V_{K+} channels; Phase 4 – initially Na+ enters then both Na+ and Ca++ (via T-type Ca++ channels) till it reaches threshold level of 40mV

SPOT 4

1. A = ANS; B = Chemical control; C = Preload; D = Contractility; E = Afterload
2. Stimulation of sympathetic NS; Hormones such as epinephrine and norepinephrine; Increased Ca++ levels in ECF; Drugs like Digitalis
3. It increases the permeability of SA node to K+ ions by slowing the closure of K+ channels → hyperpolarization → prolonging the time required to reach the threshold

SPOT 5

1. a) Capillary hydrostatic pressure, b) Interstitial fluid hydrostatic pressure, c) Plasma colloid oncotic pressure, d) Interstitial fluid osmotic pressure
2. Forces causing filtration: a, d; Forces causing reabsorption: b, c
3. Under physiological limits, the heart normally pumps all the blood returned to it.

SPOT 6

1. A = Skeletal muscle; B = Cardiac muscle
2. Skeletal muscle: Dihydropyridine rec is physically attached to Ryanodine rec and upon activation opens it releasing Ca++ from sarcoplasmic reticulum
   Cardiac muscle: Dihydropyridine rec acts as a voltage gated Ca++ channel; Entry of Ca++ acts on Ryanodine rec opening it and releasing further calcium
3. 1. Dihydropyridine rec; 2. Ryanodine rec; 3. Sarcoplasmic reticulum

SPOT 7

1. Rt ven: Sys → 25 (range 15 – 30) mmHg; Dias → 4 (range 0 – 8) mmHg
   Lt ven: Sys → 130 (range 90 – 140) mmHg; Diast → 9 (range 5 – 12) mmHg
2. Diastolic pressure + $1/3^{rd}$ of pulse pressure
3. Baroreflex, Chemoreflex, CNS ischemic response